Equations of Motion

1) Which of the following is a vector quantity?
(distance, displacement, time, mass)
Answer Hint
Displacement

2) Negative acceleration is known by what name?
Answer Hint
Retardation

3) Measurements that indicate direction along with magnitude are ----------.
Answer Hint
Scalar quantities

4) Time is a ---------- quantity. (vector/scalar)
Answer Hint
Scalar

5) The magnitude of displacement of an object can be equal or ---------- or ---------- to the distance travelled.
Answer Hint
Less than, zero

6) Magnitude of displacement and distance covered are equal only when an object is moving along a straight line in the ----------.
Answer Hint
Same direction

7) Velocity is a ---------- quantity. (vector/scalar)
Answer Hint
Vector

8) The ---------- of velocity and displacement are the same. (direction/unit)
Answer Hint
Direction

9) An object travelling along a circular path with constant speed always changes its ----------. (direction/mass)
Answer Hint
Direction

10) If the direction of motion of an object with constant speed is changing, then the ---------- of the object will change. (speed/velocity)
Answer Hint
Velocity

11) What are the quantities that can be found out from a velocity-time graph?
Answer Hint
Displacement, acceleration, velocity, time. (any two)

12) The equations of motion are applicable only for objects in motion having ----------.
Answer Hint
Constant acceleration or uniform acceleration

13) What colour clothes are preferred in situations where you have to walk on the road in dim light?
Answer Hint
Light colour

14) Write down the differences between distance and displacement.

Answer Hint

distance	displacement
Total length of the path travelled	The straight-line distance between two positions
It cannot be zero	It can be zero
Scalar	Vector

15) A person is sitting at the top of a 320 m high tower. An object fell from him. How long will it take for the person to hear the sound of the object falling from the top and hitting the floor? (g = 10 m/s 2 , speed of sound in air = 320 m/s)
Answer Hint
$s = ut + \frac{1}{2} at^2$
$u = 0, \quad s = 320 \text{ m}, \quad a = 10 \text{ m/s}^2, \quad t = ?$
$s = ut + \frac{1}{2} at^2$
$320 = 0 \times t + \frac{1}{2} 10 \times t^2$
$320 = 0 + 5\ t^2$
$t^2 = 320/5 = 64$
$t = 8$ s
That means an object falling from the top of the tower takes 8 seconds to reach the floor.
Since the speed of sound is 320 m/s, sound takes 1 s to travel 320 m from the floor.
So it takes 8+1 = 9 s for the person above to hear the sound.

16) Write the safety rules to be followed by pedestrians and motorists to ensure road safety. (two each)
Answer Hint
Pedestrian:-
If there is a footpath on the side of the road, you should travel on it.
If there is no footpath, pedestrians should walk on the right side of the road.
Motorist:-
The vehicle should be driven on the left side.
Must have a driving license.

17) What is a vector quantity? Give an example.
Answer Hint
Vector quantities are quantities that require direction along with magnitude. Vector quantities have magnitude and direction.
Example: Displacement

18) What is a scalar quantity? Give an example.
Answer Hint
Scalar quantities are quantities that do not require direction.
Example: Distance

19) If a 5 m long bus crosses a 19 m long bridge with a velocity of 3 m/s, find the time taken by the bus to cross the bridge.
Answer Hint
s = 19 + 5 = 24 m, v = 3 m/s
v = s/t
t = s/v = 24 / 3 = 8 s

20) Calculate the displacement of a car travelling at 10 m/s in 2 minutes.
Answer Hint
v = s/t
s = v x t = 10 x 2 x 60 = 1200 m

21) A bullet train travels 800 km in 2 hours. Calculate the speed of the train.
Answer Hint
v = s/t = 800 / 2 = 400 km/hr
Or
s = 800 km = 800 x 1000 = 800000 m
t = 2 hr = 2 x 60 x 60 = 7200 s
v = 800000 / 7200 = 111.11 m/s

22) Classify the following as those with uniform velocity and non-uniform velocity.
a) A train moving at constant speed in a straight line
b) Earth's motion around the Sun
c) Light travelling through the same medium
d) Motion of a car on a busy road
Answer Hint
Uniform velocity: a, c
Non-uniform velocity: b, d

23) A car travelling at 20 m/s changes its velocity to 40 m/s in 4 s. Calculate the acceleration of the car.
Answer Hint
u = 20 m/s, v = 40m/s , t = 4 s
a = (v – u)/t = (40 – 20) / 4 = 5 m/s^2

24) A car at rest started travelling in a straight line. If the acceleration of the car is 4 m/s2, what is its velocity after 10 s?
Answer Hint
u = 0, a = 4 m/s^2, t = 10 s
v = u + at = 0 + 4 x 10 = 40 m/s

25) A motorcycle travelling with a speed of 5 m/s comes to rest in 5 s. Calculate the acceleration of the bike.
Answer Hint
u = 5 m/s, v = 0 m/s , t = 5 s
a = (v -u) / t = (0 – 5) / 5 = -1 m/s^2

26) The velocity of a bus starting from rest increases to 10 m/s in 6 s. Calculate the acceleration of the bus.
Answer Hint
u = 0 m/s, v = 10 m/s , t = 6 s
a = (v -u) / t = (10 – 0)/6 = 1.66 m/s^2

27) The acceleration of a car travelling at 5 m/s is 2 m/s2 What will be the velocity of the car after 5 s?
Answer Hint
u = 5 m/s, a = 2 m/s^2, t = 5 s
v = u + at = 5 + 2 x 5 = 15 m/s

28) List the circumstances under which objects may have positive and negative acceleration.
Answer Hint: Positive acceleration:-
Falling stone
A train leaving the station
Negative acceleration:-
A stone is thrown up
Train arriving at the station

29) If a motorcycle travelling at a speed of 15 m/s decelerates by 2 m/s^2, how much time will it take for the bike to come to rest?
Answer Hint
u = 15 m/s, v = 0 m/s , a = 2 m/s^2
a = (v - u) / t
t = (v – u) / a = (0 – 15) / (-2) = 7.5 s

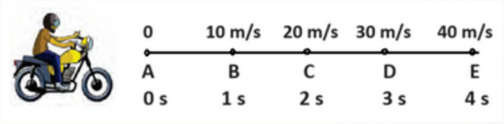
30) On the signboards installed on the roadside, warnings are given using signs like circles, triangles, squares etc. What are the warnings given by each of these?



Answer Hint
a) Mandatory signs
b) Cautionary signs
c) Informatory signs

31) A bus travelling with a speed of 20 m/s comes to rest in 10 s. Then calculate the acceleration of the bus. How much is the retardation?
Answer Hint
u = 0 m/s, v = 20 m/s , t = 10 s
a = (v -u) / t = (0 – 20) / 10 = -20 / 10 = -2 m/s^2
Retardation = 2 m/s^2

32) Given is a diagram relating the motion of a bike. Complete the velocity-time table using the information given in the figure and sketch the velocity-time graph.



Answer Hint
Constructing and completing the table.
Drawing the graph.

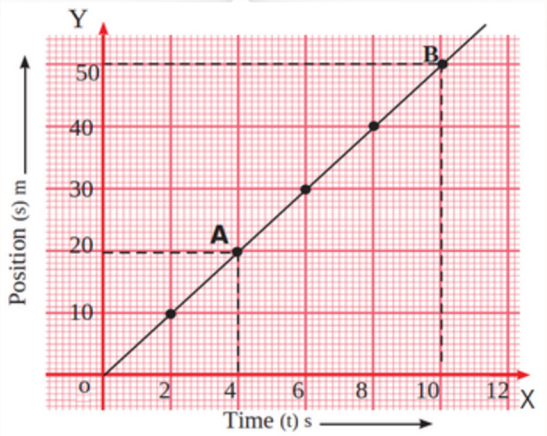
33) Observe the table.
a) Draw the graph using the information in the table
b) By what name is this graph known?
c) From the shape of the graph obtained, write the characteristic of the object's velocity.

Time (s)	2	4	6	8	10
Position (m)	5	10	15	20	25

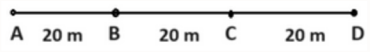
Answer Hint
a) Illustrates the graph.
b) Position–time graph
c) Uniform velocity

34) Observe the graph.
a) Find the displacement of the car in 4 s from the graph.
b) Find the velocity between 2 s and 6 s.
c) What is the time taken to travel 50 m?

Answer Hint
a) 20 m
b) 5 m/s
c) 10 s

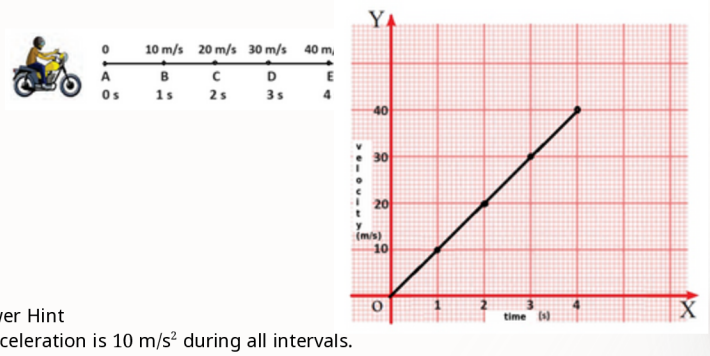


35) A child starts from A and goes to B and then to C.
a) What is the distance travelled by the child?
b) If the child then travels from C to D, does the displacement of the child make a difference? And the distance?
c) Is the displacement of the child from A through B to C positive or negative?



Answer Hint
a) Displacement = 20 m + 20 m = 40 m, Distance = 20 m + 20 m = 40 m
b) Distance and displacement will vary.
c) Positive

36) Given the graph relating the motion of a bike.
a) Find the acceleration of the bike at each interval from the velocity-time graph.
b) Is the motion of this bike uniform or non-uniform? Justify your answer.



Answer Hint
a) Acceleration is 10 m/s^2 during all intervals.
Eg: - A to B: Acceleration, a = v / t = 10 / 1 = 10 m/s^2
b) Uniform acceleration. acceleration is equal in equal intervals.

37) An object is thrown upward from a building 80 m high with a speed of 30 m/s. (g = 10 m/s 2).
a) What is the maximum height the object reaches?
b) How long will it take to reach the ground?
Answer Hint
a) u = 30 m/s, v = 0 m/s , a = -10 m/s^2
$v^2 = u^2 + 2$ as
$s = (v^2 - u^2) / 2a = (02 - 302) / (2 \times -10) = 45$ m
Maximum height = 45 m + 80 m = 125 m
b) To reach the maximum height from the top of the building
u = 30 m/s, v = 0 m/s , a = -10 m/s^2
v = u + at
$t = (v - u)/a = (0 - 30)/-10 = 3$ s
$t_1 = 3$ s
To reach the ground from the maximum height
u = 0 m/s, s = 125 m , a = 10 m/s2
$s = ut + ½ at^2$
$125 = 0 + ½ \times 10t^2$
$t^2 = 25$
$t_2 = 5$ s
Total time $t = t_1 + t_2 = 2 + 5 = 7$ s

38) A boy runs down a track 200 m long and returns to the starting position in 30 s.
a) What is the total distance travelled by the child?
b) What is the displacement of the child?
c) What is the child's speed?
d) What is the velocity of the child?
Answer Hint
a) 200 m
b) 0
c) speed = distance/time = 200/30 = 6.67 m/s
d) velocity = 0

39) a) Formulate the definition of distance and displacement.
b) In which case displacement of an object is zero?
c) In which case distance and displacement are equal?
Answer Hint
a) Writing definitions.
b) When the object returns to its original position
c) When the object travels along a straight line in the same direction

40) A person travels 100 km by car from town X to town Y. Another person travels 60 km by train on a straight track from X to Y.
a) What is the distance travelled by the car?
b) What is the displacement of the car?
c) What is the distance travelled by the train?
d) What is the displacement of the train?
Answer Hint a) 100 km b) 60 km c) 60 km d) 60 km